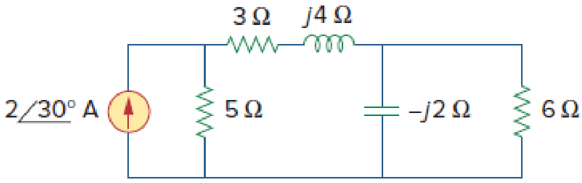
Chapter 11, Problem 56P

Problem

Obtain the complex power delivered by the source in the circuit of Fig. 11.75.

Figure 11.75:



Step-by-step solution

Step 1 of 4

Refer to Figure 11.75 in the text book.

In Figure 11.75, the impedances $-j2\ \Omega$ and $6\ \Omega$ are in parallel.

$$\begin{aligned} R_a &= -j2\ \Omega \| 6\ \Omega \\ &= \frac{(-j2)(6)}{-j2+6} \\ &= 0.6-1.8j \end{aligned}$$

In Figure 11.75, the parallel impedance is series with series combination of 3Ω and $j4\Omega$.

$$\begin{aligned} R_b &= (0.6-1.8j)+(3+4j) \\ &= (3.6+2.2j)\Omega \end{aligned}$$

Step 2 of 4

Draw the simplified circuit diagram of Figure 11.75.

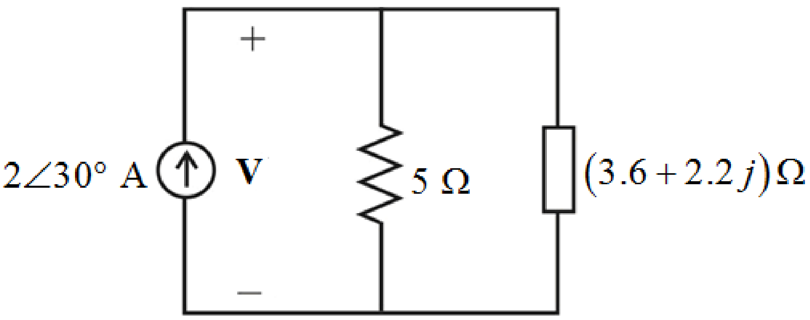


Figure 1

Step 3 of 4

In Figure 1, the impedances $(3.6 + j2.2)\ \Omega$ and $5\ \Omega$ are in parallel.

$$\begin{aligned} R_{eq} &= (3.6 + j2.2) \| 5 \\ &= \frac{5(3.6 + j2.2)}{5 + (3.6 + j2.2)} \\ &= (2.2716 + 0.6979j)\Omega \end{aligned}$$

Determine the voltage across the current source.

$$\mathbf{V} = \mathbf{I}_s (2.2716 + 0.6979j)$$

Substitute $2\angle 30^\circ\ \text{A}$ for $\mathbf{I}_s$ in the equation.

$$\begin{aligned} \mathbf{V} &= (2\angle 30^\circ)(2.2716 + 0.6979j) \\ &= 4.75\angle 47.08^\circ\ \text{V} \end{aligned}$$

Step 4 of 4

Determine the complex power delivered by the source.

$$\mathbf{S} = \frac{1}{2}\mathbf{V}\mathbf{I}_s^*$$

Substitute $4.75\angle 47.08^\circ\ \text{V}$ for $\mathbf{V}$ and $2\angle 30^\circ\ \text{A}$ for $\mathbf{I}_s$ in the equation.

$$\begin{aligned} \mathbf{S} &= \frac{1}{2}(4.75\angle 47.08^\circ)(2\angle 30^\circ)^* \\ &= \frac{1}{2}(4.75\angle 47.08^\circ)(2\angle -30^\circ) \\ &= [4.54 + 1.4j]\ \text{VA} \end{aligned}$$

Thus, the complex power delivered by the current source is, $\boxed{\mathbf{S} = [4.54 + 1.4j]\ \text{VA}}$.